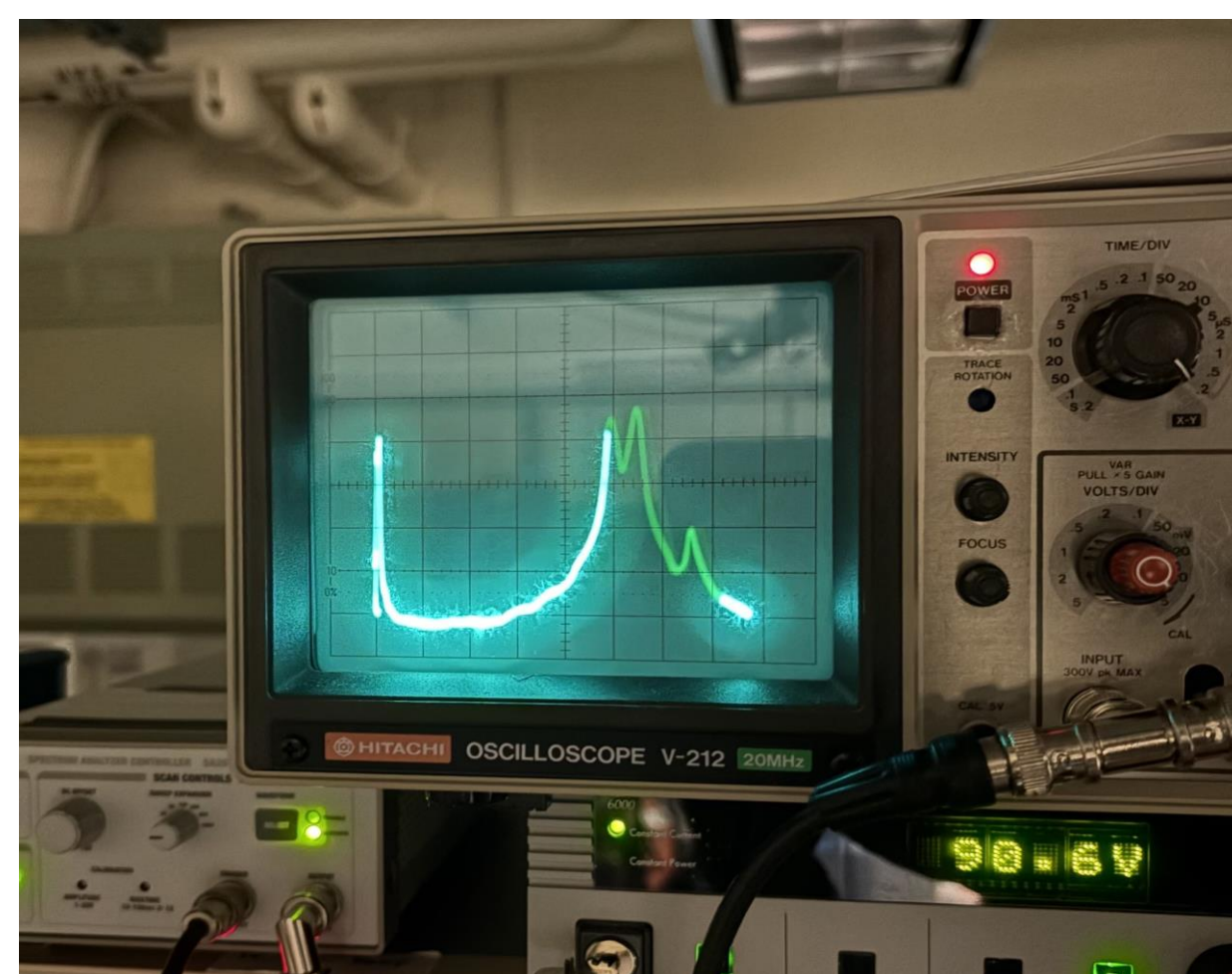


Introduction

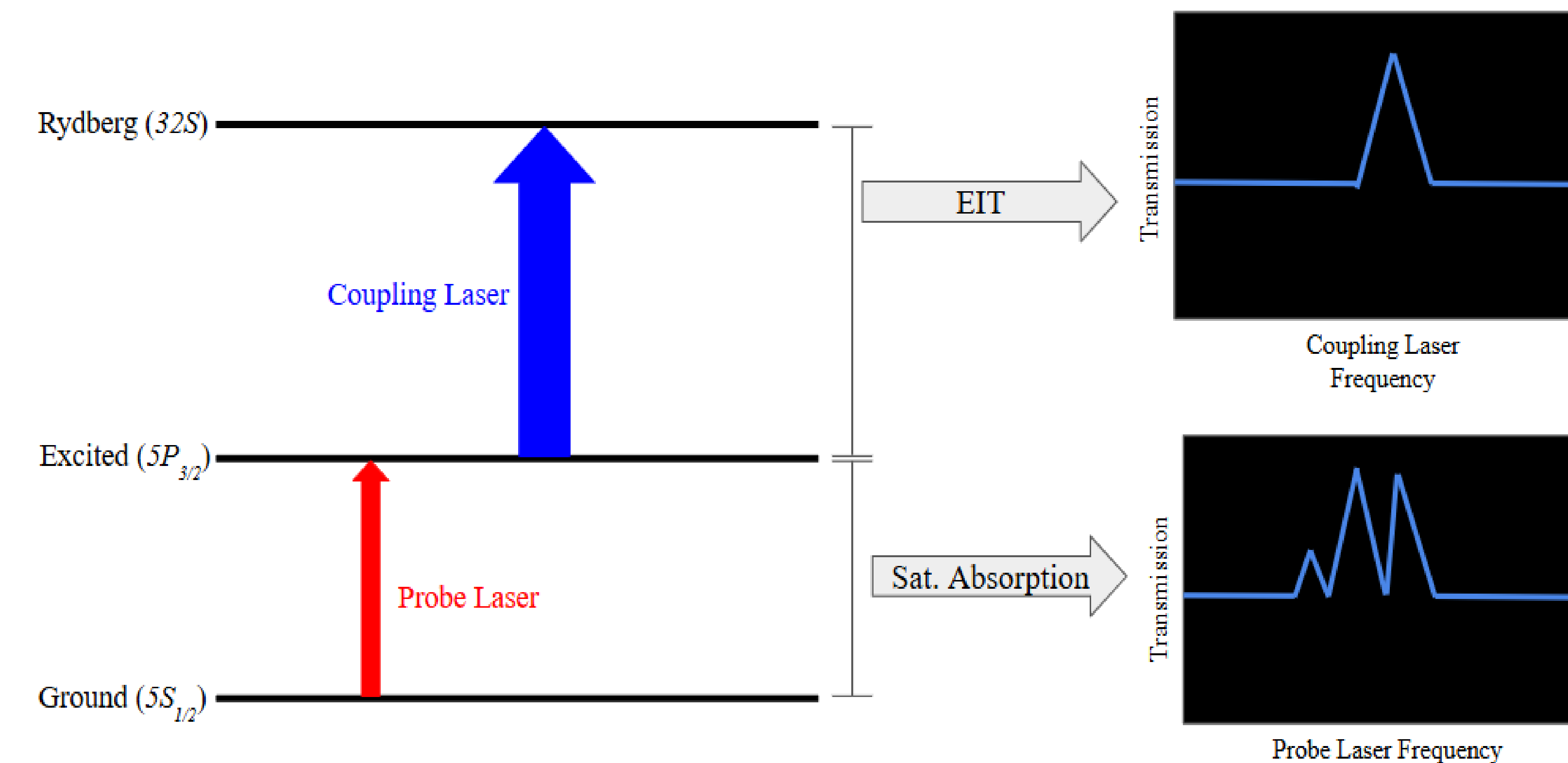
- Rydberg atomic states are characterized by $n > 10 \rightarrow$ the most probable distance between nucleus and outermost valence electron is much larger than in normal atoms.
- Can be employed in precision EM field sensing, as well as neutral-atom quantum computers.
- Rubidium atoms can be driven into Rydberg states using lasers carefully tuned to proper transition frequencies.
- This experiment utilized a two-stage process, with two voltage-tunable external cavity diode lasers (ECDLs):
 - Stage 1: Drive $5S_{1/2} \rightarrow 5P_{3/2}$ transition and lock with saturated absorption spectroscopy.
 - Stage 2: Drive $5P_{3/2} \rightarrow 32S$ (Ryberg) transition and lock with electromagnetically induced transparency (EIT) spectroscopy.

Saturated Absorption Spectroscopy

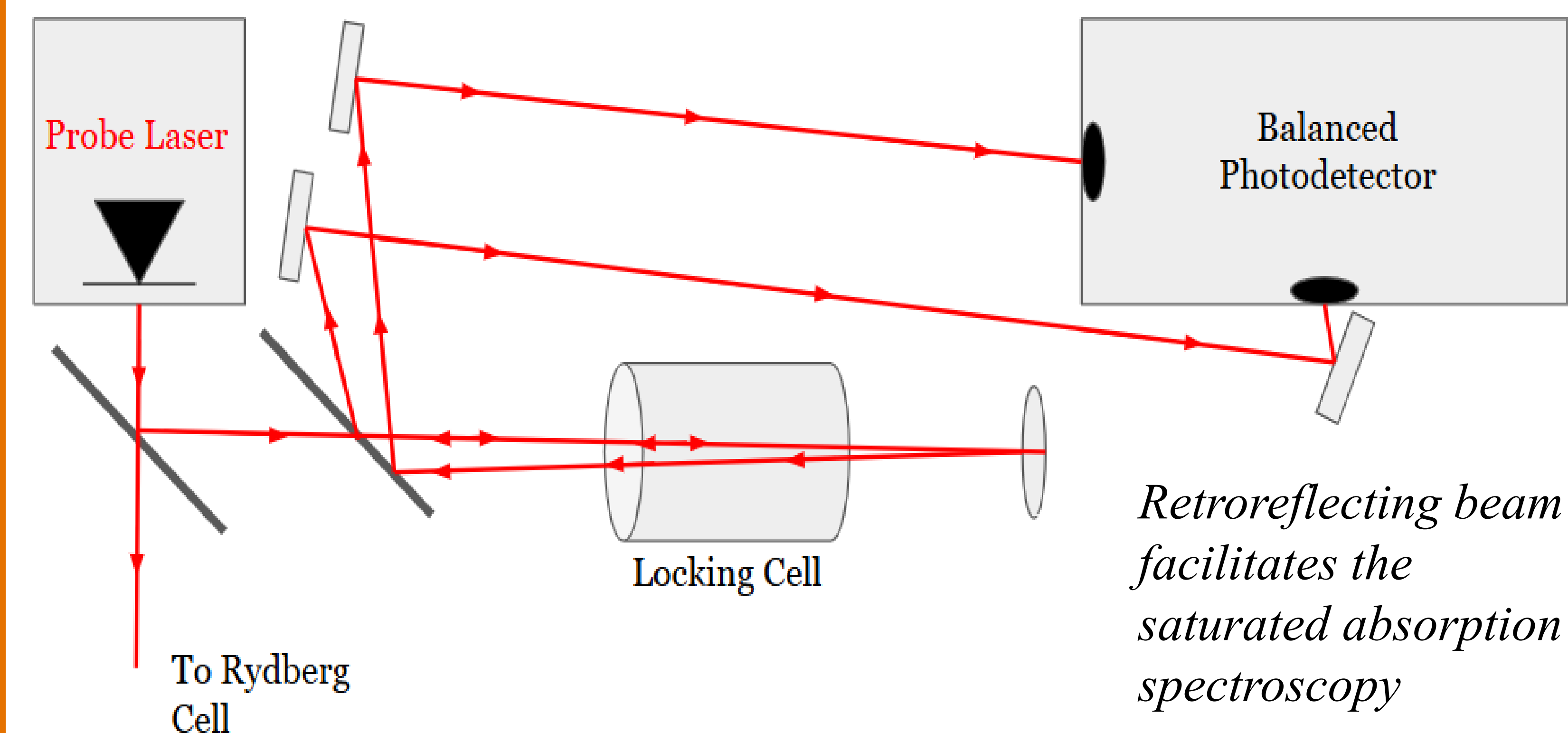


- The probe laser passes through a Rubidium filled locking cell and is then split in two with one beam retroreflecting. When the laser has the $5S_{1/2} \rightarrow 5P_{3/2}$ transition frequency, the first beam “saturates” the transition.
- In this case, the retroreflecting beam is *more* intense than the non-retroreflecting. This corresponds to a peak in laser transmission. This method also eliminates Doppler Broadening.
- Probe laser piezo voltage was scanned and PID feedback controller was used to lock to a crossover peak.
- Successful and durable probe laser lock was achieved!

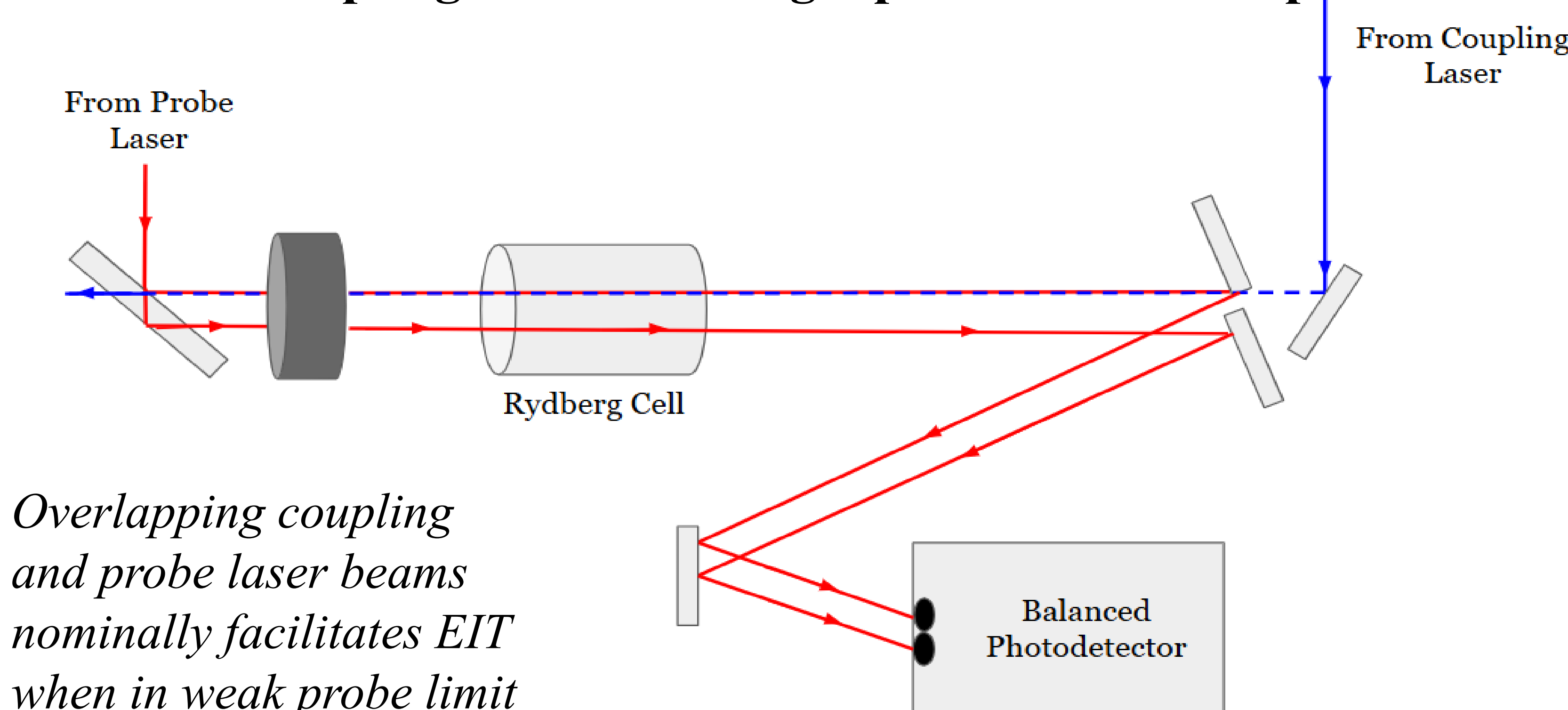
Cartoon Level Diagram and Signals



Probe Laser Locking Optical Bench Setup

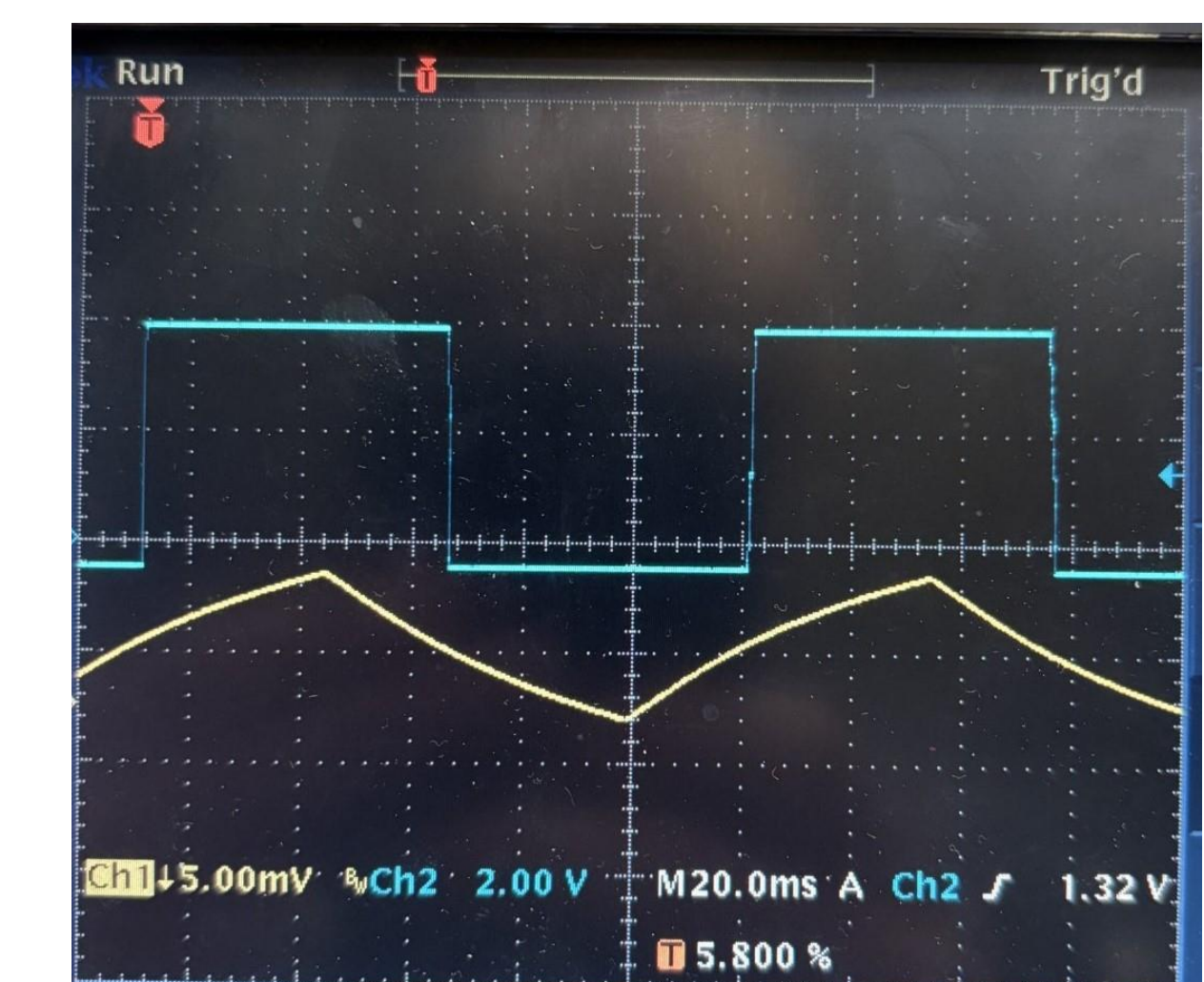


Coupling Laser Locking Optical Bench Setup



EIT Spectroscopy

- When probe laser is locked to the $5S_{1/2} \rightarrow 5P_{3/2}$ transition and a coupling laser is overlapped with it in a Rubidium cell, if coupling laser is scanned near the $5P_{3/2} \rightarrow 32S$ transition, EIT can occur.
- EIT is when in the presence of a tuned coupling laser the ground and first excited state become decoupled $\rightarrow$ leads to peak in transmission!
- In the simplified 3 level ladder Hamiltonian considered here, EIT can only occur in weak probe limit: probe laser intensity $\ll$ coupling laser intensity
- Observed a clear coupling laser dependent signal



- Yet, encountered issues observing EIT. Attempted to:
 - Boost coupling laser intensity and reduce probe laser intensity using neutral density filters
 - Isolate coupling laser signal using chopper and lock-in amplifier
 - Clear Rubidium cell of stray electrons using a small electric field
 - Improve wavelength scanning accuracy by changing wave-meters.
- None of these improvements led to EIT observation.

Conclusion

- Developed simple and effective saturated absorption spectroscopy lock for probe laser.
- Carried out extensive troubleshooting of coupling laser setup, ruled out many potential sources of error
- Future work: attempting same procedure with new Rydberg cell and/or attempting to transition to a lower n Rydberg state (less coupling laser power required)